



ENERGETIKA
ENERGETICS

Published from 2004
Ministry of Press and Information
of Azerbaijan Republic,
Registration number 3337, 07.03.2011

ISSN 1816-2126
Number 01, 2025
Section: English

ECOENERGETICS

HONORARY EDITOR IN CHIEF: Fagan G. Aliyev
SENIOR EDITOR: Rashad G. Abaszade
MANAGING EDITOR: Elmira A. Khanmamadova

INTERNATIONAL REVIEW BOARD

Arif Pashayev, Azerbaijan	Rafiq Aliyev, Azerbaijan	Turhan Vaziroglu, USA
Vagif Abbasov, Azerbaijan	Fuad Hajizadeh, Azerbaijan	Shiro Takada, Japan
Vagif Farzaliyev, Azerbaijan	İsmayil Aliyev, Azerbaijan	Luca Di Palma, İtalia
Natig Cavadov, Azerbaijan	Nazim İmanov, Azerbaijan	Yuriy Tabunshikov, Russian
Khadiyya Khalilova, Azerbaijan	Ali Guliyev, Azerbaijan	Mithat Kaya, Turkey
Farhad Aliyev, Azerbaijan	Leyla Mammadova, Azerbaijan	Elvin Aliyev, UK
Sakin Cabarov, Azerbaijan	Nazim Shamilov, Azerbaijan	Emre Gür, Turkey
Adil Azizov, Azerbaijan	Salahaddin Yusifov, Azerbaijan	Volodymyr Kotsyubynsky, Ukraine
Azer Mammadov, Azerbaijan	Mazahir İsayev, Azerbaijan	Matlab Mirzayev, Russian
Nurmammad Mammadov, Azerbaijan	Yusif Aliyev, Azerbaijan	Olga Kapush, Ukraine
Akif Alizadeh, Azerbaijan	Adil Abdullayev, Azerbaijan	Aitbek Aimukhanov, Kazakhstan
Rahim Alakbarov, Azerbaijan	Alakper Hasanov, Azerbaijan	Maksym Stetsenko, China
Gurban Eyyubov, Azerbaijan	Sevinj Malikova, Azerbaijan	Aida Bakirova, Kyrgyzstan
Samad Yusifov, Azerbaijan	Tarlan Huseynov, Azerbaijan	Bakhtyar Soltabayev, Kazakhstan
Ruslan Nuriyev, Azerbaijan	Tamella Naibova, Azerbaijan	Dzmitry Yakimchuk, Belarusiya
Agali Guliyev, Azerbaijan	Oktay Salamov, Azerbaijan	Saifullah Khalid, İndia
Tural Nagiyev, Azerbaijan	Asif Pashayev, Azerbaijan	Anitha Velu, İndia
Latif Aliyev, Azerbaijan	İlman Hasanov, Azerbaijan	Koushik Guha, India
		Nitesh Dutt, India
		Neeraj Kumar Bhoi, India
		Akbar Haghi, Portugal
		Muhammad Sultan, Pakistan

TECHNICAL EDITORIAL BOARD

SENIOR SECRETARY: İmran Y. Bayramov, Afig M. Nabiyev, Karim G. Karimov, Turan A. Nahmatova, Nigar V. Abbasova, Rashid Y. Safarov, Nuranə A. Zohrabbayli, Seynura A. Hasanova, Kamila A. Cafarli.

PUBLISHING OFFICE

5, M.Rahim, AZ-1073, Baku Azerbaijan

Tel.: 99412 538-23-70, 99412 538-40-25 Fax: 99412 538-51-22

E-mail: info@ieeacademy.org ekoenergetics@innovationresearch.az

Internet: <http://ieeacademy.org> www.innovationresearch.az

CONTENTS

1	Advances and Trends in Micro- and Nano-Fabrication Rashad Abaszade, Elmira Khanmamadova	1-23
2	Application of drones to hybrid energy systems: operating cost reduction and advantages N.T.Vahabli	24-28
3	Economic and environmental studies in the sustainable development of renewable energy in Karabakh M.Huseynova	29-32
4	Comparative analysis of oil-contaminated soil remediation methods A.Azizov, S.Huseynli	33-35
5	Development of an Automated Control System for Crude Oil Heater and Its U.Rzayev, İ.Bayramov	36-38
6	Electromagnetic waves and their effects on the human organism T. Huseynov, B. Davudov	39-42
7	Investigating the impact of climate change on the caspian sea level and ecological conditions A.Khalilova, N.Farzaliyeva	43-48
8	Laser strengthening of CP65+5%MF abrasive material for gears of oil pumps A.Guliyev, Y.Jahangirova	49-56
9	Plastic Alternatives: Bioplastics and Nanotechnology L.H. Memmedova, F.F. Ismayilova	57-59
10	Rehabilitation, modernization and expansion of electrical networks in liberated territories K.Gulmammadov	60-68
11	Renewable energy sources in the territory of east zangezur and Karabakh S.Gasimova, G.Mammadova	69-71
12	Sustainable urbanization and green infrastructure for reducing carbon emissions S.Gasimova, N.Əhədzadə	72-75
13	Solutions for optimizing heat dissipation on the printed circuit board of the drive circuit of a DC motor Jamil Gunashov, Rashad Abaszade	76-82
14	Prompt-based forecasting of monthly solar irradiation in the fuzuli district using ai models Mobil Maharramov	83-86
15	Industrial waste processing and utilization: a study of Sumgayit technopark C.A.Gurbanov	87-90
16	Artificial Intelligence and Geology F.M.Hajizadeh, N.Y.Ahadi	91-95
17	Energy Sector Vulnerability Fidan Abdullazade, Farhad Aliyev	96-99
18	Water Degradation Challenges in Azerbaijan's Karabakh Region S. R. Yusifova	100-103
19	Waste collection and utilization at sumgait technologies park C.A.Gurbanov	104-106

Plastic Alternatives: Bioplastics and Nanotechnology

L.H. Memmedova, F.F. Ismayilova

Azerbaijan University of Architecture and Construction

l.mamedova.ekologiya@gmail.com, Ismayilova.fidan.00@mail.ru

Abstract: Plastic pollution has become one of the most pressing environmental challenges of the 21st century, with billions of tons of plastic waste entering the ecosystem every year. As traditional plastics are derived from non-renewable resources and take hundreds of years to degrade, there is a growing need for sustainable alternatives. This article explores two promising solutions: bioplastics and nanotechnology. Bioplastics, made from renewable biological resources, offer a biodegradable alternative to conventional plastics, while nanotechnology can enhance the performance and environmental impact of plastic materials. Both technologies present opportunities to reduce plastic waste, but also come with challenges related to economic viability, environmental sustainability, and long-term effects. The paper concludes with insights into the future of bioplastics and nanotechnology in the global effort to combat plastic pollution.

Keywords: Bioplastics, Nanotechnology, Plastic Pollution, Sustainable Materials, Nanocomposites, Biodegradability, Renewable Resources

Corresponding authors, email: Issafura17111@outlook.com

1. Introduction

The rise of plastic products in the global market has revolutionized industries, from packaging and healthcare to automotive and construction. However, the widespread use of plastic, particularly single-use plastics, has led to significant environmental concerns. Plastics, especially those derived from petrochemicals, persist in the environment for hundreds of years, accumulating in oceans, landfills, and ecosystems. This has prompted urgent calls for more sustainable alternatives.

Bioplastics and nanotechnology represent two key areas of innovation that could offer solutions to the plastic waste crisis. Bioplastics are made from renewable biological resources such as plant starch, algae, and agricultural waste. These materials have the potential to reduce our dependence on fossil fuels and minimize plastic waste through biodegradability. On the other hand, nanotechnology involves manipulating matter at the molecular level, allowing for the creation of materials with enhanced properties, including stronger, lighter, and more durable plastics that are also more environmentally friendly.

While both bioplastics and nanotechnology offer promising alternatives, they come with their own sets of challenges. Bioplastics must be produced sustainably to avoid negative environmental impacts, and the cost of production is often higher compared to conventional plastics. Nanotechnology, while enabling enhanced plastic properties, raises concerns about toxicity and the environmental impact of nanomaterials. This article examines the potential of these technologies to address plastic pollution, their advantages, limitations, and the future of sustainable plastic materials [6,7].

Bioplastics: Eco-Friendly Alternatives

Bioplastics are materials made from renewable, plant-based resources. Unlike conventional plastics, which are derived from fossil fuels, bioplastics are biodegradable or compostable under the right conditions. The main types of bioplastics include:

- a) Polylactic Acid (PLA): PLA is made from fermented plant starch, typically derived from corn. It is widely used in food packaging and single-use items, such as cups, plates, and cutlery. PLA is compostable under industrial conditions, making it a preferable option for reducing waste in some applications.
- b) Polyhydroxyalkanoates (PHA): PHAs are produced by microorganisms from renewable carbon sources. These plastics are biodegradable in both industrial composting environments and natural ecosystems. They are used in a variety of applications, from packaging to medical devices.

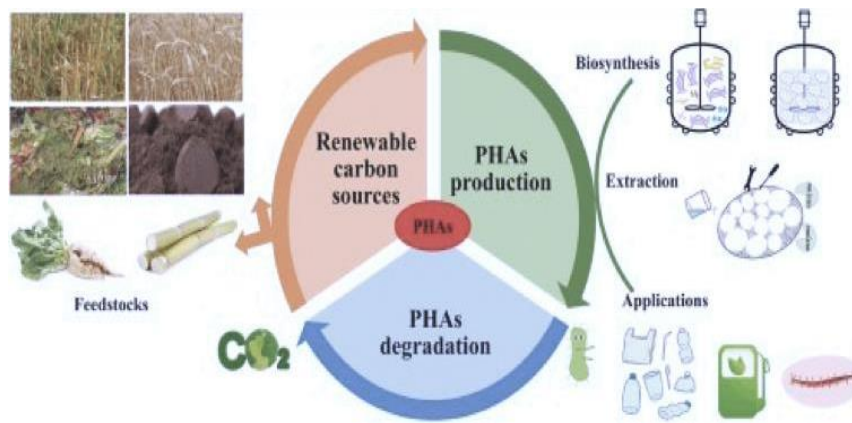


Fig 1. Polyhydroxyalkanoates (PHA) processes

- c) **Starch-Based Plastics:** These are derived from starch-rich crops such as corn or potatoes. While they are not as durable as other bioplastics, they decompose quickly and can be used in short-lived products such as packaging and bags [5].

Bioplastics offer significant environmental benefits. They are derived from renewable resources, reducing reliance on petroleum-based plastics. Additionally, many bioplastics are biodegradable, which helps to reduce the long-term environmental impact of plastic waste. However, challenges remain, particularly with the land and water use associated with large-scale production of bioplastics. Further research is needed to make bioplastic production more efficient and sustainable [2,7].

Nanotechnology in Plastics: Enhancing Performance and Sustainability

Nanotechnology involves manipulating materials at the nanoscale (one-billionth of a meter) to create new materials with improved properties. In the context of plastics, nanotechnology has the potential to create stronger, lighter, and more sustainable materials [3].

Nanocomposites: These are plastic materials that have been enhanced with nanoscale particles, such as carbon nanotubes or nanocellulose. Nanocomposites are stronger, lighter, and more durable than conventional plastics, making them ideal for industries like automotive, aerospace, and electronics. For example, carbon nanotubes improve the strength and conductivity of plastics, making them more versatile for high-performance applications.

Improved Barrier Properties: Nanoparticles can be incorporated into plastic films to enhance their resistance to moisture, oxygen, and UV radiation. This is particularly beneficial for packaging, as it can extend shelf life and reduce food waste. Additionally, nanotechnology can help reduce the amount of material required for packaging, further decreasing the environmental footprint of plastic use [6].

Biodegradability Enhancement: Nanotechnology can also be used to accelerate the biodegradation process of plastics. Nanocellulose, derived from plants, can be used to enhance the biodegradability of both bioplastics and conventional plastics, allowing them to break down faster and more completely in the environment [1].

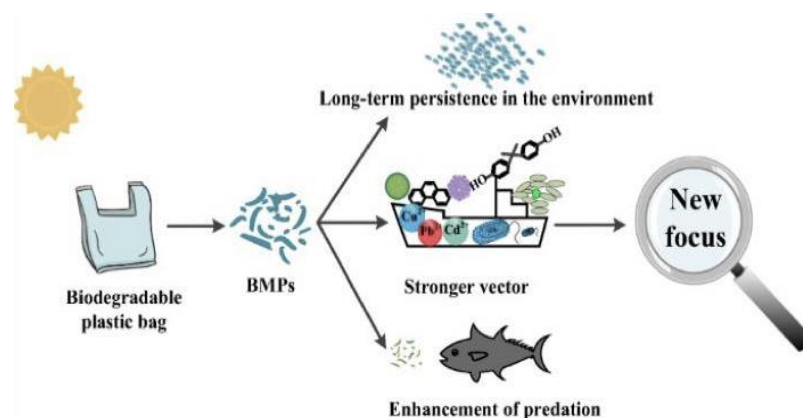


Fig 2. Example of Biodegradable plastic bag

Nanomaterials: Nanotechnology is also being researched for its potential to help clean up plastic waste from the environment. Certain nanomaterials can attract and capture plastic particles, particularly microplastics, from water or soil. This research could lead to new methods for addressing the growing problem of microplastic pollution [4].

CONCLUSION

Bioplastics and nanotechnology offer promising solutions to the global plastic pollution crisis, each with unique advantages. Bioplastics provide a renewable, biodegradable alternative to conventional plastics, but challenges related to production efficiency and sustainability remain. Nanotechnology, on the other hand, can improve the performance of plastics, reduce material usage, and enhance biodegradability, but it raises concerns about the potential toxicity of nanoparticles and their long-term effects on the environment. Despite these challenges, the continued development of bioplastics and nanotechnology holds significant promise for a more sustainable future. With further research, innovation, and improvements in production methods, these technologies could play a crucial role in reducing plastic waste and minimizing environmental harm. Governments, industries, and consumers must collaborate to support the transition to more sustainable plastic alternatives, ensuring that these technologies are adopted responsibly and effectively.

REFERENCES

1. Bioresource Technology. (2021). Biodegradable plastics and bioplastics: Current trends and future prospects. *Bioresource Technology*, 340, 1255–1263. <https://doi.org/10.1016/j.biortech.2021.1255>
2. European Bioplastics. (2020). Bioplastics market data 2020. European Bioplastics e.V. Retrieved from European Bioplastics Market Data
3. Koh, L., & Chin, K. (2019). Nanotechnology in the plastic industry: Current applications and future prospects. *Journal of Nanomaterials*, 25(2), 57–72.
4. Lange, J. P., & Goot, W. M. (2020). Nanotechnology for sustainable plastics: Innovations and future challenges. *Journal of Nanotechnology*, 26(8), 432–445. <https://doi.org/10.1016/j.jnano.2020.04.014>
5. Ravi, S., & Sengupta, A. (2020). Bioplastics: Present and future of sustainable materials. *Materials Science and Engineering Journal*, 45(7), 1123–1139.
6. Singh, J., & Kaur, A. (2022). Nanocomposites for sustainable plastics: Applications and challenges. *Nano Research Letters*, 17(6), 345–356.
7. Teli, M. D., & Rajput, N. S. (2021). Sustainability of bioplastics: From materials to applications. *Environmental Science and Technology*, 34(3), 179–195.

A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a blue sky with clouds. The image is framed by a dark green curved border at the top.

ECO

ENERGETIKA
ENERGETICS

A graphic of a water splash or wave, rendered in a soft, painterly style, located on the right side of the lower half of the cover.